Along the solutions of (1.3), the energy is conserved, since its derivative vanishes:

$$\frac{\mathrm{d}}{\mathrm{d}t}E = \sum_{j=1}^{\mathrm{N}} m_j \dot{\boldsymbol{q}}_j \cdot \ddot{\boldsymbol{q}}_j + \sum_{j=1}^{\mathrm{N}} \frac{\partial U}{\partial \boldsymbol{q}_j} \cdot \dot{\boldsymbol{q}}_j = \sum_{j=1}^{\mathrm{N}} \left(m_j \ddot{\boldsymbol{q}}_j + \frac{\partial U}{\partial \boldsymbol{q}_j} \right) \cdot \dot{\boldsymbol{q}}_j = 0.$$

As will become apparent in later chapters, it is sometimes necessary to introduce additional terms into the model to account for environmental conditions; these may destroy or alter the conservation of energy.

When we refer to the *value* of energy in a conservative system we typically write E instead of E (which represents the energy function).[3] Thus the positions and velocities of system evolving under Newtonian dynamics satisfy the equation

$$E(\boldsymbol{q}_1, \boldsymbol{q}_2, \ldots, \boldsymbol{q}_{\mathrm{N}}, \dot{\boldsymbol{q}}_1, \dot{\boldsymbol{q}}_2, \ldots, \dot{\boldsymbol{q}}_{\mathrm{N}}) = \mathrm{E}.$$

In addition to the energy, there may be other constants of motion which arise from the structure of the equations of motion. For example, if the full set of particles of an N-body system is included in the model, then the instantaneous force acting on particle i due to the presence of particle j will be the exact negative of the force acting on particle j due to particle i; this is a direct consequence of Newton's third law (every action has an equal and opposite reaction). This means that the sum of all the forces will vanish. Since

$$\frac{\mathrm{d}\boldsymbol{p}_i}{\mathrm{d}t} = \boldsymbol{F}_i,$$

we have

$$\sum_{i=1}^{\mathrm{N}} \frac{\mathrm{d}\boldsymbol{p}_i}{\mathrm{d}t} = \sum_{i=1}^{\mathrm{N}} \boldsymbol{F}_i = 0$$

and thus the three components of the *total momentum vector* $\boldsymbol{p}_{\mathrm{tot}} := \sum_{i=1}^{\mathrm{N}} \boldsymbol{p}_i$ will be conserved quantities. Additional first integrals sometimes arise in specific situations, depending on the form of the potential energy function.

For a few very special N-body systems, analytical solution is possible.

Example 1.3 (Harmonic Oscillator) Consider the system

$$\dot{x} = v, \qquad \dot{v} = -\Omega^2 x.$$

[3] We use roman (non-italic) letters for parameters that describe the thermodynamic state, e.g. number of degrees of freedom, the volume, the energy, the temperature or the pressure.